KUKU — AI GENERALIST ASSIGNMENT

Problem 1: Find the Promos (Ads) | Srikrishna Valepe | Technical Architecture & Solution Document

Live Prototype: https://kuku-promo-detector.onrender.com	**Code Repository:** https://github.com/srikrishna6174/kuku-ai-promo-detector	**Test Suite:** 20/20 Passing Tests (OK)

1. Executive Summary & End-to-End System Workflow

Microdramas represent a high-throughput vertical video format where editorial teams manually scrub 50+ episode timelines to isolate high-converting 20–60s hooks for TikTok, Instagram Reels, and YouTube Shorts. Our production platform eliminates manual scrubbing by converting raw, unstandardized screenplays into verified, editor-ready timestamp cut sheets with zero hallucination risk.

Step-by-Step Architecture:

1. Multi-Format Ingestion: A universal ingestion engine processes `.txt` screenplays, subtitle streams (`.srt`, `.vtt`), raw dialogue transcripts, or bulk `.zip` archives. It extracts dialogue blocks, tone indicators, sound cues, and normalizes timestamps into precise millisecond offsets.

2. Deterministic Window Generation: Generates candidate cut windows bound strictly between 20.0s and 60.0s (target 25–45s). To prevent unnatural abrupt cuts where characters are cut off mid-reaction, an automatic +0.75s post-dialogue visual buffer is applied.

3. Multi-Tier Hybrid Detection: Candidate windows pass through: (a) *Deterministic Rules* (0–3s hook disruption, conflict progression, cliffhanger acoustic/visual peaks); (b) *Generative AI* (semantic tension, emotional payload, and human-readable editorial rationale); (c) *Multi-LLM Failover* with Gemini and OpenAI adapters.

4. Human-in-the-Loop Studio: Editors inspect collapsible promo tiers (Manual Cuts, AI-Generated, Rule-Based), pin custom timestamp adjustments, review trailer stitches, and export NLE markers (CSV) and lossless FFmpeg stream-copy scripts without baked video overlays.

2. Architectural Division: Deterministic Rules vs. Generative AI

Dimension	Deterministic Code & Pattern Rules	Generative AI / LLM Layer
Scope & Responsibilities	• Screenplay parsing (.txt, .srt, .vtt, .zip) & timestamp math. • Sliding window generation (20s–60s bounds, +0.75s buffer). • Anchor detection: Dialogue starts, high-intensity cues, verbs. • Mathematical scoring of 0–3s hook & final cliffhanger peak.	• Semantic tension analysis, nuance & subtext extraction. • 'Why it works' rationale for editors and marketing leads. • LLM-as-a-Judge validation pass across candidate pairs. • Pluggable REST adapter (Gemini & OpenAI auto-failover).
Speed & Cost	Offline-capable on CPU; parses 50 episodes in <1.8s at $0 cost.	Evaluates top candidate windows (~2–4/ep) on demand.

3. Scoring Rubric & Window Selection Logic

Candidate clips are evaluated on a 100-point composite rubric across three quantitative dimensions:

• **Hook Disruption Score (40%):** Evaluates the opening 3.0 seconds. High weight assigned to confrontation verbs (*'marry', 'turn around', 'betray', 'enough'*), high-voltage acoustic stings (*slap, door bursts*), or urgent dialogue cues that arrest vertical feed scrolling.

• **Cliffhanger Peak Score (35%):** Evaluates the terminal 3.0–5.0 seconds. Identifies unresolved questions, suspended dialogue ellipsis (*'...'*), dramatic tonal swells, or hard visual cut-to-black beats that compel viewers to tap through to the full episode.

• **Standalone Narrative Coherence (25%):** Requires >= 2 distinct characters in conflict within 4–10 dialogue turns. Penalizes monologues and heavy exposition that require prior backstory knowledge.

4. Strategic Campaign Rollout: Funnel-Aligned Publishing Framework

Microdrama marketing requires matching narrative tension with user acquisition funnel stages rather than posting by raw episode order:

Campaign Tier	Target Episodes	Strategic Rationale & Acquisition Purpose
Tier 1: Launch Promos Cold Audience UA	**Episodes 1 – 3** (e.g., Ep 3: 00:25–00:56 Ep 1: 01:15–01:38)	**Zero-Lore Universality:** The wedding betrayal, public humilation, and unexpected hero intervention require zero context. Maximum conversion from casual scrollers to new app installs.
Tier 2: Escalation Viral Growth & Shares	**Episodes 4 – 15** (Identity reveals & social reversals)	**Power Reversals & Catharsis:** The female lead stands her ground; the Duke's true status surfaces. Generates comment velocity, user debate, and organic algorithmic reach on TikTok and Reels.
Tier 3: Retargeting Churn Prevention	**Episodes 16 – 50** (Mid-season crises & ultimatums)	**High-Stakes Binge Triggers:** Life-or-death ultimatums and deep cliffhangers targeted at installed users who stalled at early episodes to re-engage them into paid binge sessions.

5. Systematic Quality Audit & Failure Modes ('Where does the tool pick badly?')

A complete audit (*src/quality_checker.py*) across all generated promo candidates identified 4 systematic failure modes:
• **1. Premature Cut Buffer (58% of raw clips):** Script dialogue ends at exact timestamps (e.g. 00:32.40). Cutting immediately clips facial reactions and lingering sound design. *Mitigation:* Deterministic +0.75s post-dialogue visual reaction padding.
• **2. Off-Target Secondary Characters (32% of raw clips):** Heated arguments between maids or side nobles score high lexically but fail as cold UA hooks. *Mitigation:* Lead-character entity recognition and prioritization.
• **3. Exposition & Lore Overload:** High-word-count exchanges (>25 words per line) introduce too much backstory. *Mitigation:* Turn density penalty.
• **4. Visual-Only Action Ambiguity:** Intense physical actions with minimal dialogue (slaps, escapes) are under-scored by text analysis. *Mitigation:* Human editor pinning and future multimodal video support.

6. AI Engineering Tools & Key Lessons: Where AI Failed

• **AI Tools Employed:** Antigravity agent for end-to-end Python/Streamlit code synthesis, test authoring, and PDF layout; Gemini 2.5 and GPT-4o-mini REST APIs for semantic analysis, emotional arc validation, and LLM-as-a-Judge benchmarking.
• **Where Generative AI Got It Wrong:** Relying solely on LLM prompt engineering to pick timestamps resulted in three recurring flaws: (1) *Timestamp Drift:* LLMs frequently hallucinated timestamps outside exact subtitle boundaries; (2) *Duration Violations:* Prompts failed to enforce strict 20s–60s bounds, outputting 12s or 85s clips; (3) *Mid-Sentence Audio Clipping:* Truncating on the exact final token without reaction buffers. *Resolution:* Constraining LLMs within deterministic sliding window geometry and 20 automated unit tests resolved these failures completely.

7. Production Scalability & Deliverables (Moments Only — No Artificial Overlays)

• **Scalability:** Deterministic rule filtering processes 1,000+ episode scripts in <45s on CPU, keeping cloud costs minimal by reserving LLM calls for top-scoring candidate windows. Editors export timestamped cut sheets without artificial overlays or burned-in subtitles.
• **Final Deliverables:** (1) **Live Deployed Prototype:** https://kuku-promo-detector.onrender.com; (2) **Source Repository:** https://github.com/srikrishna6174/kuku-ai-promo-detector; (3) **Structured Output Package:** *outputs/promos/* (JSON, CSV, individual text cut sheets, stitched trailers), *outputs/quality_audit/*, and *outputs/editor_exports/* (Premiere markers & FFmpeg scripts); (4) 20/20 automated unit tests.